Huawei Access Node Restoration Procedure for Hardware/Environmental Faults (ALARM_4519)

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4519

Description

The Huawei access node, designated as BAS-ACC-73, situated at the Basavanagudi POP, is susceptible to hardware or environmental faults, manifesting as ASIC memory parity errors, line card crashes, or power supply redundancy loss. This failure domain is precipitated by the intrinsic complexities of the access node's architecture, wherein the line cards, equipped with Application-Specific Integrated Circuits (ASICs), are responsible for packet processing and forwarding. The ASICs, being susceptible to thermal fluctuations and voltage irregularities, can experience memory parity errors, leading to line card crashes. Furthermore, the power supply redundancy loss can occur due to failures in the power modules or the power distribution units, thereby compromising the overall system availability. The Huawei CLI dialect is utilized to diagnose and rectify these faults, leveraging the vendor's proprietary commands to navigate the device's control and data planes.

Pre-requisite Checks

- Dispatch remote hands to Basavanagudi POP to perform on-site checks
- Verify redundant link status to ensure no single point of failure
- Check environmental parameters such as temperature and humidity

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to thermal fluctuations	High
Line card crashes resulting from voltage irregularities	Medium
Power supply redundancy loss due to power module failure	Low

Diagnostic Commands & Expected Outputs

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Command: display diagnostic-information, Expected Output: Diagnostic information, including ASIC memory parity error counts and line card crash logs, Description: This command isolates the control plane, providing insights into the device's diagnostic information, including ASIC memory parity error counts and line card crash logs, thereby facilitating the identification of the root cause of the failure.

Command: display ip interface brief, Expected Output: IP interface status, including link up/down state and packet statistics, Description: This command probes the data plane, displaying the IP interface status, including link up/down state and packet statistics, which aids in determining the impact of the failure on the network traffic.
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Resolution Steps

Step 1: Step: Replace faulty line card, Command: replace board 0/1, Description: This step involves replacing the faulty line card with a spare one, ensuring that the replacement card is properly seated and secured, and then executing the 'replace board' command to notify the system of the change.

Step 2: Step: Restore power supply redundancy, Command: power supply redundancy enable, Description: This step entails restoring power supply redundancy by enabling the redundant power supply modules, thereby ensuring that the system can continue to operate even in the event of a power module failure.

Post-Check Verification

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Command: display diagnostic-information, Expected Output: Updated diagnostic information, indicating clearance of ASIC memory parity errors and line card crashes, Description: This command verifies that the diagnostic information has been updated, indicating that the ASIC memory parity errors and line card crashes have been cleared, thereby confirming the effectiveness of the resolution steps.

Command: display ip interface brief, Expected Output: Updated IP interface status, indicating stable link state and normal packet statistics, Description: This command verifies that the IP interface status has been updated, indicating a stable link state and normal packet statistics, thereby confirming that the network traffic has been restored to normal.
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Roll-back Procedures

- Step: Revert to original line card configuration, Command: cancel replace board 0/1, Description: This step involves reverting to the original line card configuration by canceling the replacement operation, ensuring that the system is restored to its pre-failure state.

- Step: Disable power supply redundancy, Command: power supply redundancy disable, Description: This step entails disabling power supply redundancy, which should only be performed if the restoration of redundancy fails, as it may compromise system availability.